CS430/910: Foundations of Data Analytics

Social Network Analysis 1 | Ayse Sunar

Social Networks

- "**Social Networks**" refers to the pattern of interactions formed between individuals
  - E.g. the social network of a university, an office, a sports club...
  - Social Networks studied by sociologists for decades:
    - Milgram's "six degrees of separation" experiments (1960s)
    - Zachary's karate club study (1970): links between 34 club members
    - Dunbar's number: ~150 (1990s)
  - Around 15-20 years ago, something changed...

Milgram, S., 1967. The small world problem. *Psychology today*, *2*(1), pp.60-67.
Zachary, W.W., 1977. An information flow model for conflict and fission in small groups. *Journal of anthropological research*, *33*(4), pp.452-473.
Dunbar, R.I., 1992. Neocortex size as a constraint on group size in primates. *Journal of human evolution*, *22*(6), pp.469-493.

Social Networking Sites

- There have been various popular online social networking sites:
  - Friendster (2002-11)
  - Myspace (2003-Present)
  - LinkedIn (2003-Present)
  - Orkut (2004-14)
  - Facebook (2004-Present)
  - Twitter (2006-Present)
  - Google+ (2011-2019)
  - TikTok (2016-Present)
  - Vine (2013-2017)

Social Networking Sites

- Allowed users to make their social network explicit:
  - Represented within computer systems
  - Quickly grew to millions of users, hundreds of millions of links/ties
  - Social Network Analysis is now primarily a computational study

Other Social Networks

- Electronic social network analysis predates online social networking sites
  - Email traffic analysed for understanding businesses
    - The famous “enron dataset” from 2004 (hundreds of thousands of emails from Enron, used for social network and online communication research)
  - World-wide web social network data analysis
    - International World Wide Web Conference since 1994
  - Blog data analysis: LiveJournal, Blogger, WordPress...
  - Citation data from scientific papers

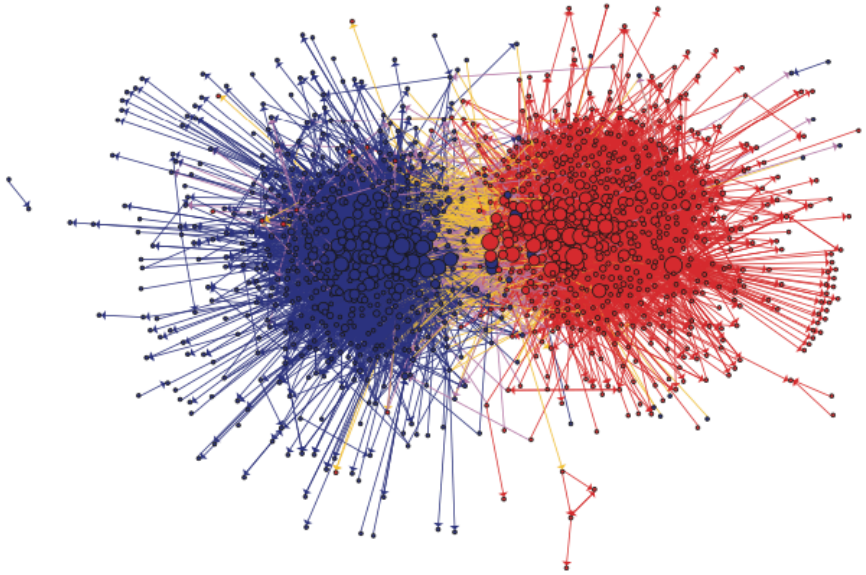
Social Networks vs. Social Media

- Aside: **social networks** distinct from **social media**
  - Social media: text (tweets), comments, photos, videos
  - Social media is often shared via social networks

Social Networks

A Graph Model

- Social networks can be modelled as a graph.
- Image to the right (from [1]):
  - Points = Political blogs in the United States
  - Red for conservatives, blue for liberals.
  - Paths between them = hyperlinks between them.
  - Red path = conservative to conservative link, blue path = liberal to liberal link, yellow path = liberal to conservative link, pink path = conservative to liberal link.

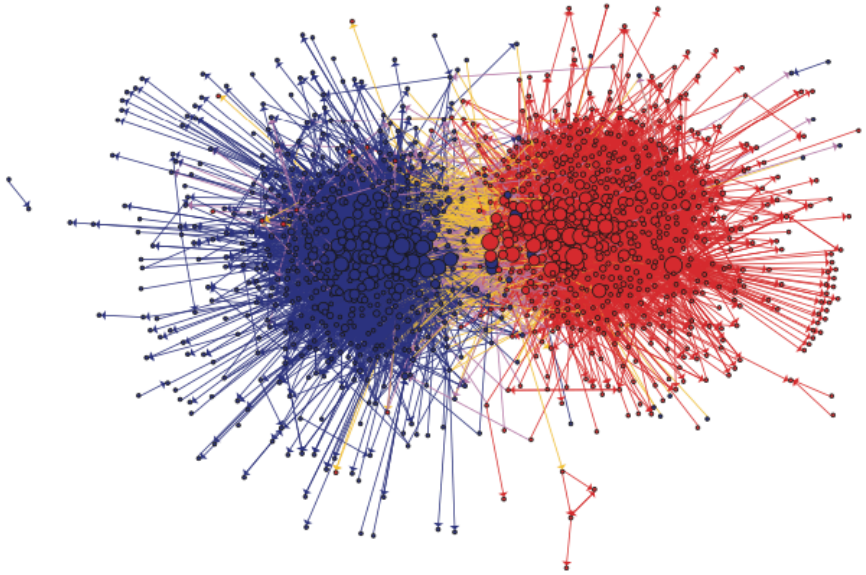


[1]Adamic, L.A. and Glance, N., 2005, August. The political blogosphere and the 2004 US election: divided they blog. In *Proceedings of the 3rd international workshop on Link discovery* (pp. 36-43).

Social Networks

A Graph Model

- Each individual entity (e.g. a user, or a blog) is a **node** (vertex)
- **Ties** (links, arcs, edges) between nodes indicate some sort of connection. E.g.:
  - A URL hyperlink connection
  - An explicit *friendship* link on Facebook (symmetric/undirected)
  - A *follow* relation on Twitter (directed)



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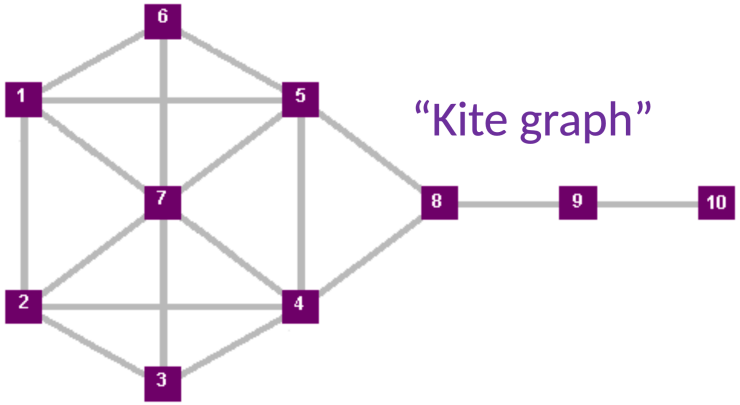
Working with Social Network Data

- Social network data is fairly widely available to study:
  - Stanford Large Datasets Collection (http://snap.stanford.edu/data/index.html), curated by Jure Leskovec
- Collect your own:
  - Twitter API, but limit on what can be sampled
  - Several social networks let you pay for access to data
  - Many social networks try to detect and limit data scraping

Social Networks

- Concepts from graph theory help understand social networks:
  - **Node Degree**: the number of ties that a node has
  - **In-Degree (Out-Degree)**: the number of directed ties pointing in (out)
  - **Distance**: minimum number of hops between a given pair of nodes
  - **Diameter**: the maximum distance between any pair of nodes
  - **Neighbours**: Nodes which are connected to a given node.



"Kite graph"

- In the above example:
  - Degree of node (7) is 6
  - Distance(1,8) is 2 (1 -> 5 -> 8)
  - Diameter is 4, realised by (1,10) (1 -> 5 -> 8 -> 9 -> 10)

Representing Social Networks

- Social network data consists of at least two data sets:
  1. A regular data set, called the *nodelist*, where each row represents an entity
  2. A data set that defines the relationships between the these entities
     - This may come in two forms, an *adjacency matrix*, or an *edgelist*
     - Adjacency matrices can be transformed into edgelists, and vice-versa

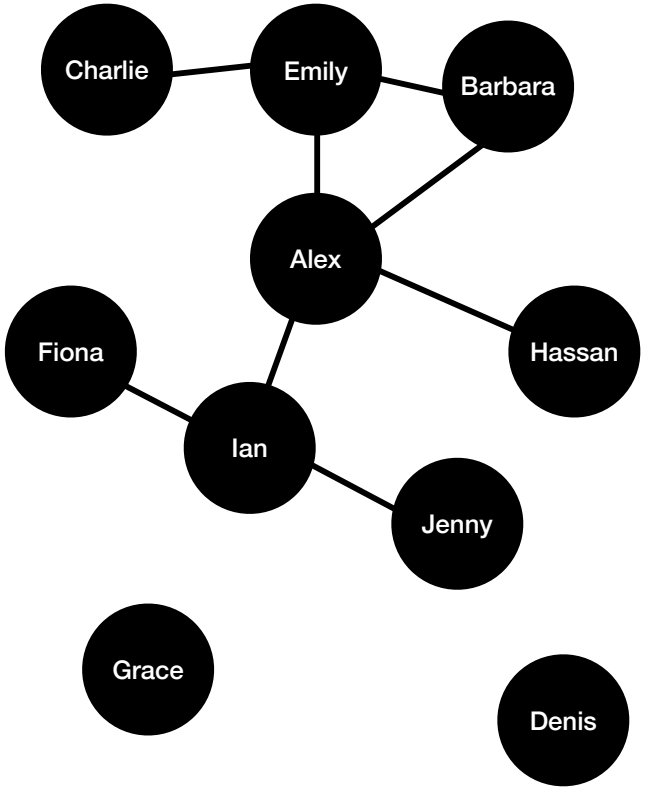
Representing Social Networks

A	Alex	Male	Russian
B	Barbara	Female	US
C	Charlie	Male	US
D	Denis	Male	Russian
E	Emily	Female	German
F	Fiona	Female	British
G	Grace	Female	Russian
H	Hassan	Male	German
I	Ian	Male	Russian
J	Jenny	Female	British

Nodelist

	A	B	C	D	E	F	G	H	I	J
A	0	1	0	0	1	0	0	1	1	0
B	1	0	0	0	1	0	0	0	0	0
C	0	0	0	0	1	0	0	0	0	0
D	0	0	0	0	0	0	0	0	0	0
E	1	1	1	0	0	0	0	0	0	0
F	0	0	0	0	0	0	0	0	1	0
G	0	0	0	0	0	0	0	0	0	0
H	1	0	0	0	0	0	0	0	0	0
I	1	0	0	0	0	1	0	0	0	1
J	0	0	0	0	0	0	0	0	1	0

Adjacency Matrix



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Representing Social Networks

A	Alex	Male	Russian
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C	Charlie	Male	US
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H	Hassan	Male	German
I	Ian	Male	Russian
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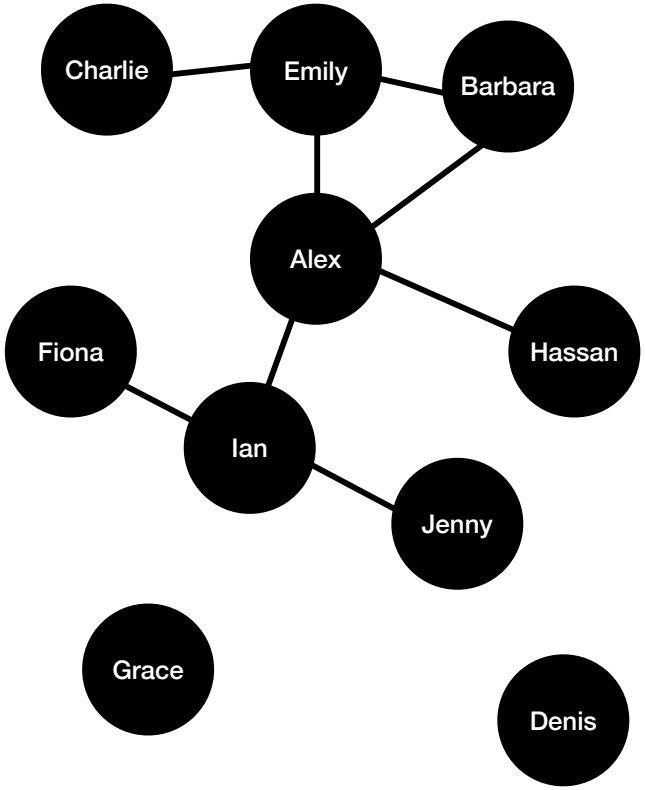
Nodelist

A	B
A	E
A	H
A	I
E	B
E	C
I	F
I	J

Edgelist (Undirected)



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Representing Social Networks

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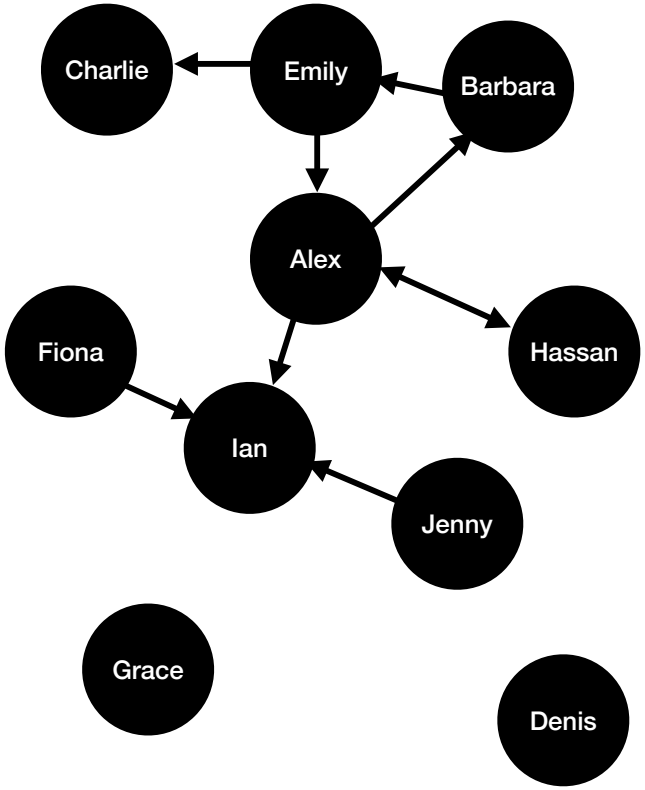
Nodelist

Source	Target
A	B
A	H
A	I
B	E
E	A
E	C
F	I
H	A
J	I

Edgelist (Directed)



Representing Social Networks

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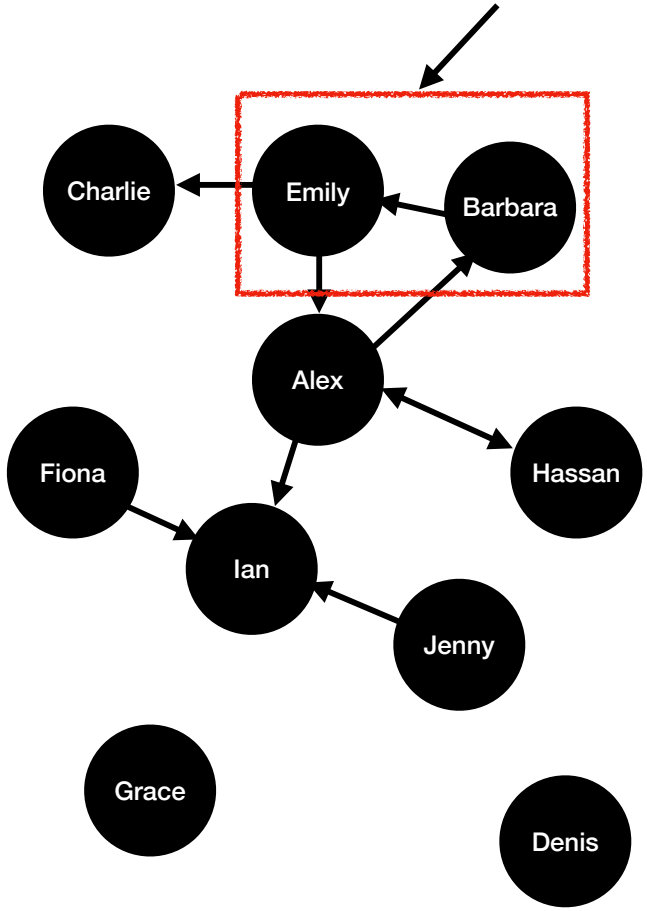
Nodelist

Source	Target
A	B
A	H
A	I
B	E
E	A
E	C
F	I
H	A
J	I

Edgelist (Directed)

Barbara is the sender or source, and Emily is the receiver or target.



Representing Social Networks

An adjacency matrix can still be used here, just it would not represent symmetric relationships!

A	Alex	Male	Russian
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C	Charlie	Male	US
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Nodelist

Source	Target
A	B
A	H
A	I
B	E
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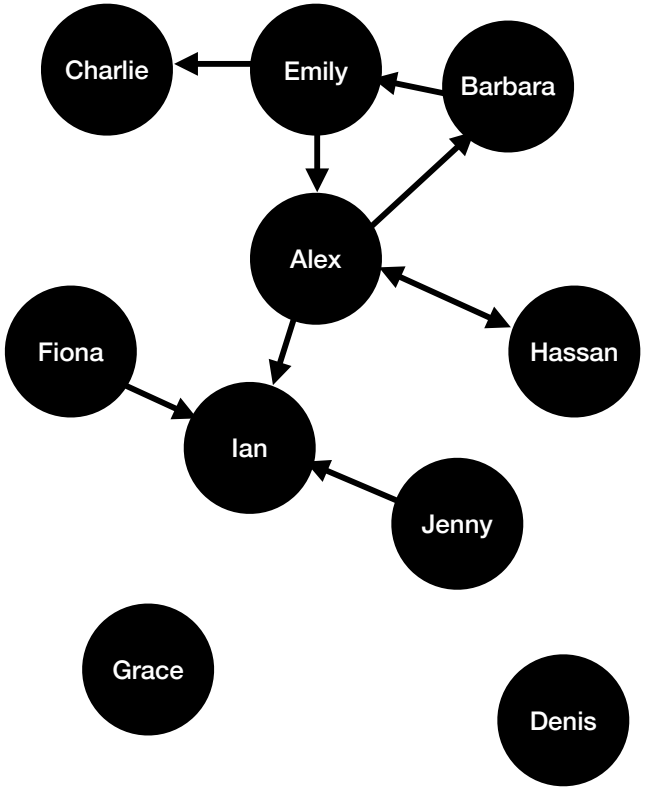
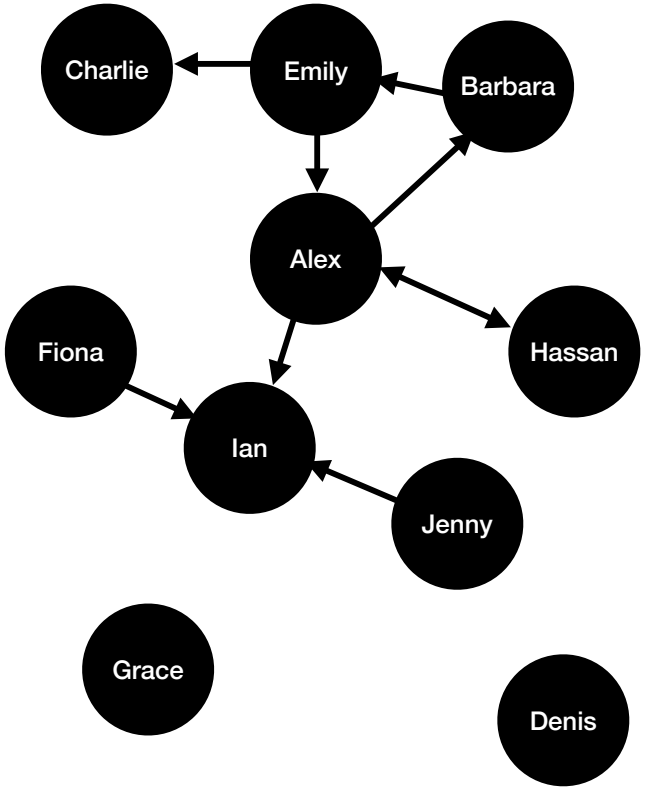
Edgelist (Directed)



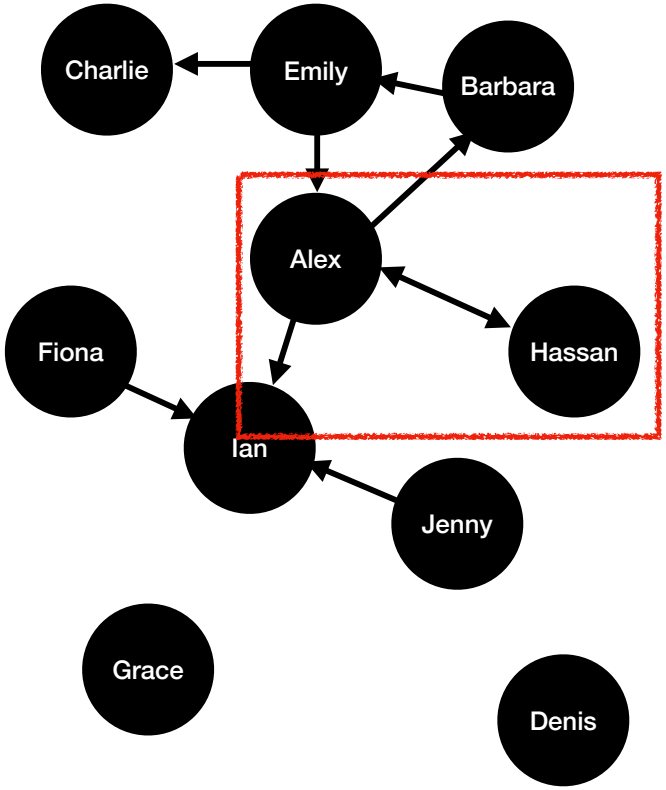
Representing Social Networks

- Consider a situation where Alex considers Hassan a friend, and Hassan shares this feeling.
- This relationship can be represented as an undirected relationship.
- However, the same graph may have a directed relationship (e.g. if Alex considers Barbara a friend, but Barbara does not consider Alex a friend).
- Rather than combining directed/undirected relationships, it is common to only use directed.



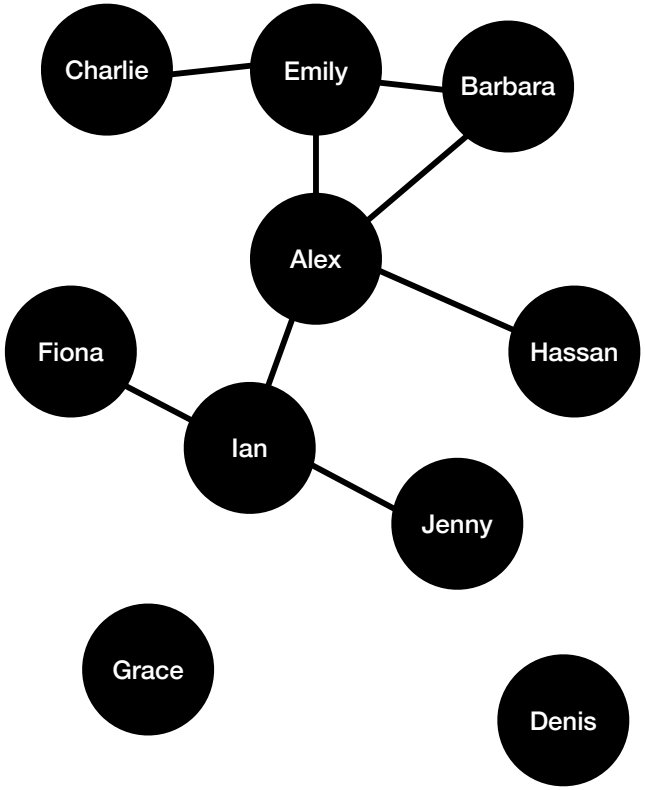
Representing Social Networks

- If there is a tie between A and B, and B and A, in a directed graph, this is known as a **reciprocated tie**.
- There is one in the figure on the right between Alex and Hassan.



Clustering Coefficient

- Most social networks look different to "(uniformly) random graphs"
- Much more likely to see a tie between a pair of neighbouring nodes than by chance
  - These are called triangles (i.e. two of my friends are also friends)
- Applies to undirected graphs, but there have been variations created which can apply to directed graphs.



Clustering Coefficient

- Let's define a graph $G = (N, A)$, where N is the set of nodes, and A is the set of ties.
- $S_i \subseteq N$ is the set of nodes connected to node $i \in N$.
- Let c_i be the cardinality of S_i.
- There are $\binom{c_i}{2}$ possible ties between nodes in S_i.
- The local clustering coefficient $e(i)$ of node i is the fraction of these possible ties that have a tie between them:

$$e(i) = \frac{\text{Number of triangles}}{\text{Number of pairs of common neighbours}} = \frac{|\{(j,k) \in A : j \in S_i, k \in S_i\}|}{\binom{c_i}{2}}$$

Aggarwal, C.C., 2015. Data mining: the textbook. Springer.

Clustering Coefficient

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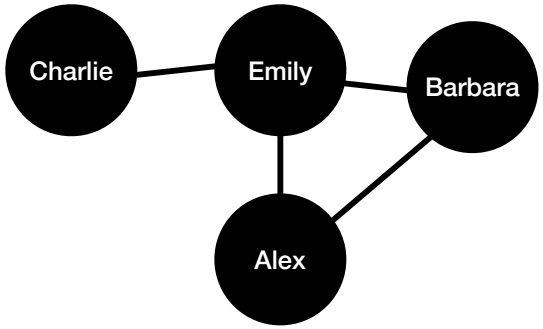
$$e(i) = \frac{\text{Number of triangles}}{\text{Number of pairs of common neighbours}} = \frac{|\{(j,k) \in A : j \in S_i, k \in S_i\}|}{\binom{c_i}{2}}$$

- The Watts-Strogatz network average clustering coefficient[1] is the average value of $e(i)$ over all nodes in the graph.

Clustering Coefficient

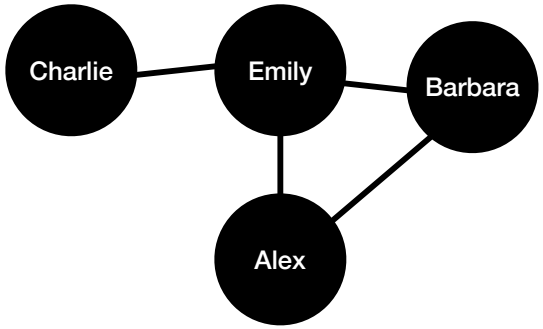


- $N = \{\text{Charlie, Emily, Barbara, Alex}\}$
- $A = \{(\text{Charlie, Emily}), (\text{Emily, Barbara}), (\text{Emily, Alex}), (\text{Barbara, Alex})\}$
- $S_{\text{Alex}} = \{\text{Emily, Barbara}\}$
- $c_{\text{Alex}} = |S_{\text{Alex}}| = 2$

$$e(\text{Alex}) = \frac{\text{Number of triangles}}{\text{Number of pairs of common neighbours}} = \frac{|\{(j,k) \in A : j \in S_{\text{Alex}}, k \in S_{\text{Alex}}\}|}{\binom{c_{\text{Alex}}}{2}}$$

$$= \frac{|(\text{Emily, Barbara})|}{\binom{2}{2}} = \frac{1}{1} = 1$$

Clustering Coefficient



- $N = \{\text{Charlie, Emily, Barbara, Alex}\}$
- $A = \{(\text{Charlie, Emily}), (\text{Emily, Barbara}), (\text{Emily, Alex}), (\text{Barbara, Alex})\}$
- $S_{\text{Emily}} = \{\text{Charlie, Alex, Barbara}\}$
- $c_{\text{Emily}} = |S_{\text{Emily}}| = 3$

$$e(\text{Emily}) = \frac{\text{Number of triangles}}{\text{Number of pairs of common neighbours}} = \frac{|\{(j,k) \in A : j \in S_{\text{Emily}}, k \in S_{\text{Emily}}\}|}{\binom{c_{\text{Emily}}}{2}}$$

$$= \frac{|(\text{Barbara, Alex})|}{\binom{3}{2}} = \frac{1}{3}$$

Social Network Analysis

- (Online) Social Networks are a topic of much study
- Inherent interest: what are they, how do they work?
- As a substitute for human behaviour
- To try to influence activity within them (advertising)
- To study activities transacted through networks
  - Look at influence on finance, news, …
  - Spot new trends, topics about to “go viral”

N.S.A. Gathers Data on Social Connections of U.S. Citizens

850

By James Risen and Laura Poitras

Sept. 28, 2013

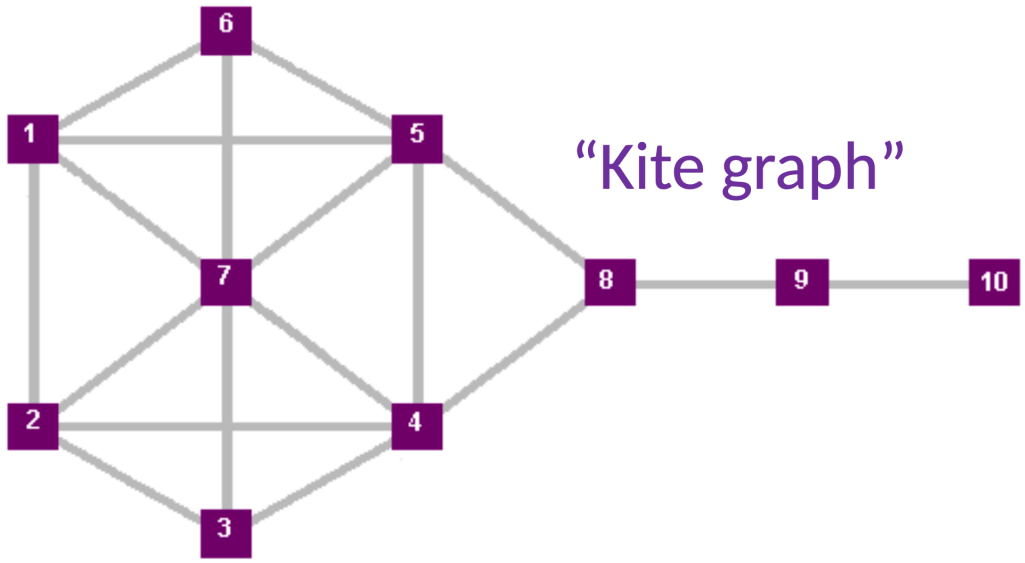
WASHINGTON — Since 2010, the National Security Agency has been exploiting its huge collections of data to create sophisticated graphs of some Americans’ social connections that can identify their associates, their locations at certain times, their traveling companions and other personal information, according to newly disclosed documents and interviews with officials.

Finding Important Nodes

- Much work on finding important or influential nodes
  - Degree: find node with highest degree
  - Not always meaningful: can engineer nodes with high degree
  - E.g. could plant nodes which all follow/friend each other
- Other more complicated measures can be used
  - E.g. betweenness centrality.

Betweenness Centrality

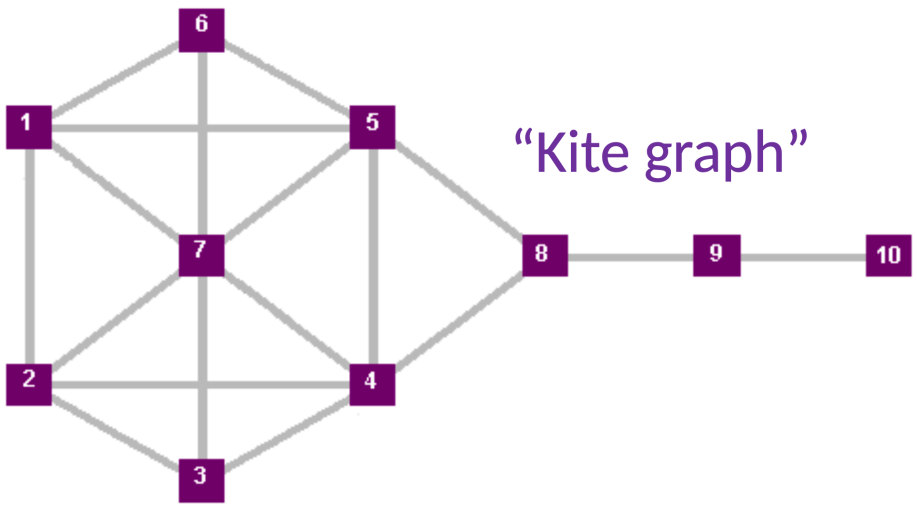
- *Betweenness*: fraction of shortest paths passing through a node
  - For all node pairs, compute the shortest path(s) between them (e.g. with Dijkstra's algorithm)
  - For each node, count the number of shortest paths it is on
  - In the kite graph, #8 is on many shortest paths



"Kite graph"

Eccentricity

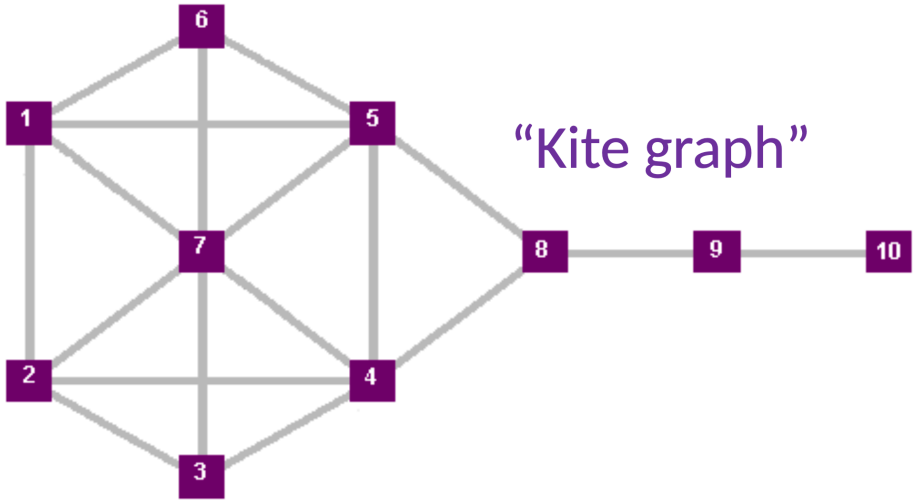
- The *eccentricity* of a node is the maximum distance to any other node in the graph
  - Eccentricity of #5 is 3 (realised by (5,10))
- If we calculate the eccentricity for all nodes:
  - The **minimum** eccentricity is known as the *radius*.
  - The **maximum** eccentricity is known as the *diameter*.



"Kite graph"

Eccentricity

- A *central node* is one that achieves the minimum eccentricity
  - #8 is the only central node here, with an eccentricity of 2



"Kite graph"

End of Module Feedback Form on Moodle

Anonymous Feedback